SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA1703

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 4 **Total: Marks:40**

Annexure A

Code of Conduct:

I E.Ragul / 192571032 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
E.Ragul

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Answer:

(a) Inductive Learning Approach, Target Variable and Key Features

Inductive learning is a supervised learning approach in which a model examines a set of labelled historical examples and derives a general rule or hypothesis that can be applied to new, unseen cases. This approach is appropriate for the diabetes dataset because every patient record already carries a known outcome (diabetic or non-diabetic), so the algorithm can learn the statistical relationship between the patient attributes and the diagnosis, and later use this learned rule to predict the status of new patients. A Decision Tree or a probability-based inductive learner is well suited here because the features are a mix of numeric and categorical clinical values, and the outcome required is a clear class label that clinicians can interpret.

The target variable is the Diabetes Status, a binary label that takes the value "Diabetic" or "Non-Diabetic" for every patient. This is the attribute the model is trained to predict.

Three key features and their justification are:

- Glucose Level – Fasting or post-prandial glucose is the primary clinical marker used by doctors to diagnose diabetes, so it carries the strongest direct signal for the target variable.
- BMI (Body Mass Index) – Excess body weight is closely linked to insulin resistance; a high BMI is one of the most consistent lifestyle-related predictors of Type 2 diabetes.

- Family History – Diabetes has a strong hereditary component, so patients with diabetic parents or siblings carry inherited genetic risk that a purely lifestyle-based feature cannot capture.

Age, blood pressure and cholesterol are also useful supporting features since insulin sensitivity decreases with age and metabolic syndrome often links high blood pressure, poor lipid profile and diabetes together, but glucose, BMI and family history are the strongest and most independent predictors and therefore form the core of the inductive model.

(b) Handling Class Imbalance

In a real hospital dataset, non-diabetic patients usually greatly outnumber diabetic patients. If this imbalance is not corrected, the model tends to become biased toward the majority class and may achieve a high overall accuracy simply by predicting "non-diabetic" for almost everyone, which is dangerous in a medical context because it increases the risk of missing true diabetic cases (false negatives).

A combination of the following strategies is recommended:

- SMOTE (Synthetic Minority Over-sampling Technique) – This is preferred over simple duplication because it generates new synthetic diabetic samples by interpolating between existing minority-class neighbours, which increases the diversity of the minority class instead of just repeating identical records, reducing the risk of overfitting.
- Class weighting – Assigning a higher misclassification penalty to the diabetic class during model training (for example, in Decision Trees or Logistic Regression) forces the algorithm to pay more attention to correctly identifying diabetic patients without altering the original data distribution.
- Under-sampling – This can be used cautiously, but only if the dataset is large enough, because randomly removing majority-class records can discard useful information.

For this medical use-case, a combination of SMOTE and class weighting is justified because it balances the training data while retaining almost all real information, giving a good trade-off between recall (catching true diabetics) and overall reliability.

(c) Dataset Split and Preprocessing

The dataset is divided into 70% for training and 30% for testing. This ratio is justified for a medical dataset because it leaves enough records for the model to learn robust patterns while still reserving a reasonably large, unseen portion to evaluate how the model performs on new patients, which is critical before any clinical deployment. Stratified splitting is used so that the proportion of diabetic and non-diabetic cases remains consistent in both the training and test sets, preventing the evaluation from being skewed by chance imbalance.

The following preprocessing steps are applied:

- Normalisation (Min-Max or Z-score scaling) – Applied to continuous features such as glucose, BMI, cholesterol and blood pressure so that all numeric attributes fall on a comparable scale; this prevents features with naturally larger numeric ranges (such as cholesterol) from dominating features with smaller ranges (such as BMI) during distance-based or gradient-based learning.
- Encoding – Categorical attributes such as gender or family history (Yes/No) are converted using label encoding or one-hot encoding so that the algorithm, which requires numeric input, can process them correctly.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

<u>Answer:</u>

(a) Decision Tree Construction and Root Node Justification

A Decision Tree classifier is built using the preprocessed training data, with Glucose Level, BMI and Family History as the primary splitting attributes. Each internal node of the tree tests one attribute against a threshold, and each branch leads either to another test or to a leaf node that predicts Diabetic or Non-Diabetic. The first three levels of the tree can be represented as follows:

- Level 1 (Root): Glucose Level <= 125 mg/dL ? -> Yes branch leads toward Non-Diabetic; No branch leads toward Diabetic.
- Level 2 (under "Yes"): BMI <= 27 ? -> Yes leads toward Non-Diabetic (low risk); No leads to further check on Family History.
- Level 2 (under "No"): Family History = Yes ? -> Yes strongly leads toward Diabetic; No leads to further check on Age.
- Level 3 (example leaves): Age <= 45 combined with moderate BMI leads toward Non-Diabetic, while Age > 45 combined with high BMI leads toward Diabetic.

Glucose Level is selected as the root node because it produces the highest Information Gain (equivalently, the lowest weighted Gini Index) among all candidate attributes. Intuitively, splitting on glucose produces child nodes that are far more "pure" (dominated by one class) than splitting on, for example, age first, because glucose is the most direct physiological marker of diabetes. A typical calculation would show the parent Gini Index close to 0.45–0.50 (near-balanced classes after SMOTE), while splitting on glucose reduces the weighted child Gini Index to around 0.20–0.25, giving the largest reduction in impurity of all candidate features, which is why the algorithm selects it first.

(b) Model Evaluation

On the 30% held-out test set, the Decision Tree is evaluated using a confusion matrix, from which the following metrics are derived: Accuracy (overall proportion of correct predictions), Precision (proportion of predicted diabetic cases that are truly diabetic), Recall or Sensitivity (proportion of actual diabetic cases correctly identified), and F1-Score (harmonic mean of Precision and Recall). A representative result for this dataset would be approximately Accuracy 82%, Precision 78%, Recall 74%, and F1-Score 76%.

False Negatives are the diabetic patients the model incorrectly labels as non-diabetic; these are the most clinically dangerous errors because such patients would not receive timely treatment. To reduce

False Negatives, the decision threshold for classifying a patient as "Diabetic" can be lowered so the model is more sensitive, and class weighting.

(c) Pruning and Clinical Deployment

An unpruned Decision Tree tends to overfit the training data by growing very deep, memorising noise instead of learning generalisable patterns. Cost-complexity (post) pruning addresses this by growing the full tree first and then removing branches whose removal does not significantly increase the overall error, controlled by a complexity parameter (alpha). Pre-pruning, in contrast, stops the tree from growing early using criteria such as maximum depth or minimum samples per leaf.

In practice, pre-pruned trees may slightly underfit and miss useful patterns, giving a training-like accuracy of about 84% but a test accuracy closer to 79% because some genuine relationships were cut off too early. Post-pruned trees usually retain more of the informative structure before trimming, achieving a test accuracy near 83%, only marginally below the unpruned tree's training accuracy, while being far more stable on new data than the unpruned tree, which may show a large gap between very high training accuracy and much lower test accuracy.

For deployment in a clinical setting, the post-pruned tree is the more appropriate choice because it offers a better balance between generalisation and interpretability: it remains simple enough for a physician to read the decision path for a specific patient, while avoiding the instability and overfitting risk of the fully grown tree, which is essential when the output directly influences a patient's diagnosis and treatment.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

<u>Answer:</u>

(a) Statistical Learning Model and Comparison with the Decision Tree

Logistic Regression is applied as the statistical learning model on the same preprocessed dataset. Unlike the Decision Tree, which partitions the feature space into rectangular regions, Logistic Regression estimates the probability of diabetes as a smooth function of a weighted linear combination of the input features, passed through a sigmoid function. This produces a continuous risk score between 0 and 1 for each patient rather than a hard split, which is useful for risk stratification (for example, categorising patients as low, medium, or high risk).

When compared on the same test set, Logistic Regression typically achieves an Accuracy of around 80% and an AUC-ROC of about 0.85, while the Decision Tree achieves an Accuracy of around 82% with an AUC-ROC of about 0.83. This shows that the two models perform similarly overall, but the statistical model provides a slightly better-calibrated ranking of patients by risk (a higher AUC-ROC), whereas the Decision Tree provides slightly higher raw accuracy and much clearer, rule-based explanations.

A statistical model such as Logistic Regression is preferable when the goal is to produce a continuous, well-calibrated probability of risk for stratifying patients into risk bands, when the relationship between features and outcome is expected to be roughly linear or additive, and when the model needs to be summarised through simple coefficients for statistical reporting. A Decision Tree is preferable when clinicians need an easily traceable, rule-based explanation for an individual decision, or when the true relationships in the data are highly non-linear with strong feature interactions that a linear model cannot capture well.

(b) Feature Importance Analysis

From the Decision Tree, feature importance is measured by how much each attribute reduces impurity (Gini Index) across all the splits where it is used. From Logistic Regression, feature importance is inferred from the magnitude and significance of the standardised regression coefficients, since a larger absolute coefficient (after scaling) indicates a stronger influence on the predicted log-odds of diabetes.

For this dataset, the top three predictors identified by the Decision Tree are typically Glucose Level, BMI and Family History, in that order, because these attributes produce the largest impurity reductions across the tree. The top three predictors identified by Logistic Regression are typically Glucose Level, BMI and Age, since these carry the largest standardised coefficients.

The two models agree strongly on Glucose Level and BMI as the two most important predictors, which is reassuring because it shows the finding is not an artefact of one particular algorithm but a genuine pattern in the data, consistent with established medical knowledge that blood sugar level and body weight are the leading indicators of diabetes risk. The two models differ slightly on the third predictor: the Decision Tree favours Family History because it produces cleanly separated branches when combined with the other splits, while Logistic Regression favours Age because it contributes a steadily increasing linear risk that is captured well by a coefficient. This difference is expected and reflects the different ways the two algorithms represent relationships (rule-based partitioning versus linear weighting), rather than a contradiction, so both Family History and Age should be reported to the clinical team as secondary but meaningful risk factors.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

Answer:

(a) MDP Formulation for Treatment Recommendation

The treatment recommendation problem is modelled as a Markov Decision Process (MDP) with the following components:

- States (S) – Each state represents a patient's current health stage, summarised from key indicators such as glucose level, BMI and blood pressure into discrete stages, for example: Stable, At-Risk, Deteriorating and Critical. This is justified because treatment decisions depend on the patient's present clinical condition, and a discretised health stage keeps the state space small enough for practical Q-Learning while still being clinically meaningful.

- Actions (A) – The available actions are Diet, Exercise, Medication and Monitor. These are justified because they represent the realistic set of interventions a clinician can prescribe at each review point, ranging from conservative lifestyle adjustment (Diet, Exercise, Monitor) to active pharmacological intervention (Medication).

- Reward Function (R) – The reward is defined as a health improvement score, calculated from the change in the patient's clinical indicators between consecutive visits; for example, a positive reward is given when glucose and BMI move closer to normal ranges after an action, a small negative reward (cost) is given for more invasive actions such as Medication to discourage unnecessary escalation, and a larger negative reward is given if the patient's condition worsens or reaches a Critical stage. This is justified because it aligns the agent's objective directly with patient wellbeing rather than an unrelated proxy.

- Transition Dynamics (P) – This defines the probability that a patient moves from one health stage to another after a given action is applied, estimated from historical patient records that show how similar patients responded to similar treatments in the past. This is justified because patient response to treatment is not fully predictable, so a probabilistic transition model captures the real-world uncertainty in treatment outcomes.

(b) Q-Learning Training

The Q-Learning agent is trained across at least five simulated patient episodes, where each episode follows one patient's health-state trajectory over several review visits. At each step, the agent chooses an action using an epsilon-greedy policy (mostly exploiting the best known action, occasionally exploring a random action), observes the resulting reward and next state, and updates its Q-table using the rule Q(s,a) <- Q(s,a) + alpha * [r + gamma * max Q(s',a') - Q(s,a)], where alpha is the learning rate and gamma is the discount factor.

An illustrative update for three state-action pairs across two iterations is shown below:

- (At-Risk, Diet): Iteration 1 updates Q from 0.00 to 0.30 after a positive reward of 2 and a modest future estimate; Iteration 2 updates Q from 0.30 to 0.48 as the estimate strengthens with more evidence.

- (Deteriorating, Medication): Iteration 1 updates Q from 0.00 to 0.55 after a larger positive reward of 4 for successfully stabilising a worsening patient; Iteration 2 updates Q from 0.55 to 0.68 as the agent gains confidence in this action for this state.

- (Stable, Monitor): Iteration 1 updates Q from 0.00 to 0.20 after a small positive reward of 1 for correctly avoiding unnecessary intervention; Iteration 2 updates Q from 0.20 to 0.27 as the value is reinforced across episodes.

The convergence criterion used is that training stops once the maximum change in any Q-value between consecutive full passes over all episodes falls below a small threshold (for example, 0.01), or once a fixed maximum number of episodes is reached, whichever comes first; at this point the Q-table is considered stable and the policy it induces is no longer changing significantly.

(c) Final Policy and Ethical Considerations

The final learned policy is extracted by choosing, for every state, the action with the highest Q-value; for example, the policy may recommend Diet or Exercise for the Stable and At-Risk stages, Medication for the Deteriorating stage, and a combination of Medication with close Monitoring for the Critical stage.

Deploying such a Reinforcement Learning agent in medical treatment recommendation raises several important ethical considerations. Patient safety must come first: the agent's recommendation should always be treated as a decision-support suggestion reviewed and approved by a qualified clinician rather than an autonomous prescription, since an incorrect action in a rare or unusual patient state could cause real harm. Bias is a serious concern because the historical data used to estimate transitions and rewards may under-represent certain patient groups (by age, gender or ethnicity), which could cause the agent to learn policies that work well for the majority but poorly for minority groups, so the training data and resulting policy must be regularly audited for fairness across subgroups. Explainability is also essential: because Q-values alone do not explain why an action was recommended, the system should be paired with interpretable summaries (for example, showing the expected health improvement and the key state indicators that drove the recommendation) so that clinicians and patients can understand and trust the reasoning before it is acted upon.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation